



FileVault

Developer & User Guide

A fast, private Android file manager built with Flutter.
Browse, organise and clean up storage – and keep the files that
matter in an encrypted Secure Folder behind an App Lock.

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1. About this release

This guide documents FileVault after the October 2026 upgrade. It is written for two readers: **developers** who build, maintain and extend the app (chapters 2–9), and **people using the app** (chapters 10–17). Chapter 18 answers common questions.

What changed in this release

Area	Change
Build errors	Root cause: a second, abandoned architecture (freezed/drift/equatable models, a duplicate operation queue, a counter-template widget test, a second l10n pipeline and a duplicate <i>com.example</i> MainActivity) lived next to the real app. None of it was reachable from <i>main.dart</i> and its packages were never in pubspec. It was removed; nothing in the shipping app depended on it. The project now needs no code generation . The missing Kotlin Gradle plugin was added.
Speed	Folder listing runs in a worker isolate with synchronous I/O (one pass instead of one async stat per file), folder item counts are computed in one background pass, revisited folders paint instantly from an in-memory cache, natural sorting no longer allocates a RegExp per comparison, copies use native <i>File.copy</i> (kernel sendfile) or 4 MB slices, same-volume moves are instant renames that skip the pre-scan, and progress updates are throttled.
Secure Folder	Native AES-256-GCM on Android (10–50× faster than the old pure-Dart cipher – what makes large videos practical), asks for PIN/fingerprint on every entry, folders can be secured, originals are removed from MediaStore and from the thumbnail cache, screenshots are blocked, decrypted previews never reach "Recent files", transfers survive the vault locking.
App Lock	New: separate 6-digit app PIN + optional fingerprint, configurable re-lock delay, brute-force cooldown, content hidden from screenshots and the recent-apps view.
UI	New palette and design tokens, white button labels in dark mode, real app icon everywhere (launcher, adaptive and themed icons, Android 12 splash, in-app), splash that waits for data, redesigned Home, lock screens and progress card, duplicate profile icon removed, many overflow fixes.
Features	Clipboard copy/cut → paste, Copy to.../Move to..., batch rename, undo for trash, hide a folder from galleries, global progress card with %, file count, speed and time left, automatic trash purge.

Verification note. The upgrade was prepared in an environment where the Flutter SDK, Android SDK and pub.dev could not be downloaded, so it could not be compiled or run there. Every change was reviewed line by line against the real package and framework sources, and by independent review passes. Please run the commands in section 4.3 once on your machine; section 4.4 lists what to do if the analyzer reports anything.

2. Architecture

FileVault is a single-module Flutter app using **MVVM with Riverpod**. The layers depend strictly inwards:

Layer	Responsibility	Rules
features/	Screens (views), their view models (Riverpod Notifiers) and screen-specific widgets.	Views hold no business logic. View models never import dart:io.
domain/	Plain models (FileEntry, VaultItem, FileOperation...) and repository interfaces.	No Flutter, no dart:io, no packages with platform code.
data/	Repository implementations, the SQLite database and services: file system, archives, vault crypto (Dart + native bridge), platform channel, thumbnails, App Lock credentials.	The only layer that touches the file system, SQLite and platform channels.
core/	DI (providers), routing, theme and design tokens, shared widgets, utilities.	Shared by all.
android/	Kotlin: platform channel (volumes, thumbnails, APK info, MediaStore sync, FLAG_SECURE), native vault crypto, foreground service for long operations.	

Data flow

A view watches a provider → the view model calls repositories → repositories use services → heavy work runs in worker isolates (listing, hashing, archiving, indexing, Dart crypto) or in Kotlin on a background thread (native crypto, MediaStore). Results come back as immutable state; the view rebuilds. Long-running file operations go through one serial queue (*OperationsController* → *OperationRunner*) that reports progress, supports pause/cancel and drives the notification.

State management

- **Riverpod 2.6** *Notifier/NotifierProvider* for every screen, *Provider* for services and repositories (composition root: *lib/core/di/providers.dart*).
- The database and settings are loaded in *main()* and injected with *ProviderScope* overrides, so repositories can be synchronous.
- *operationFinishedProvider* emits once per finished operation; screens listen to it to refresh.

Persistence

- **SQLite (sqflite)** – favourites, recents, tags, trash records, search index, hash cache, operation history, vault records. The file system is always the source of truth.
- **SharedPreferences** – user settings (theme, sort, App Lock options...).
- **flutter_secure_storage** (Android Keystore) – App Lock PIN hash and salt; the vault key copy used for fingerprint unlock.
- **App-private files** – encrypted vault blobs and *meta.json* under *getApplicationSupportDirectory()/vault*.

3. Folder structure

```

lib/
  main.dart          opens DB + settings, runs FileVaultApp
  app.dart           MaterialApp.router, lifecycle (App Lock, vault
                    auto-lock), lock overlay, global progress card

  core/
    constants/       AppConstants (paths, prefs keys, limits)
    di/              providers.dart (composition root), app_state.dart
    errors/          Failure (sealed) + Result<T>
    router/          go_router config + route names
    theme/           AppColors, AppTheme + FvTokens, typography,
                    category colours

    utils/           IsolateWorker, LifecycleGuard, ui_overlays
                    (popup tracker, bottom-bar reservations), formatters

    widgets/         FvAppBar/FvAppIcon, FvCard, buttons, tiles, sheets

  domain/
    models/          FileEntry, FileOperation, VaultItem, AppSettings...
    repositories/    interfaces

  data/
    database/        AppDatabase (sqlite schema)
    repositories/    *_repository_impl.dart
    services/        file_system_service, operation_runner,
                    vault_crypto_service, native_vault_crypto,
                    app_lock_service, directory_cache,
                    thumbnail_service, platform_channel_service,
                    archive_service, permission_data_source

  features/
    home/ browser/ category/ search/ analyzer/ trash/ archive/
    collections/ viewer/ vault/ security/ settings/ onboarding/ operations/
    l10n/app_localizations.dart every user-facing string
  android/app/src/main/kotlin/com/filevault/app/
    MainActivity.kt   FlutterFragmentActivity, registers channels
    FileVaultChannel.kt volumes, thumbnails, APK info, MediaStore, FLAG_SECURE
    VaultCryptoChannel.kt native AES-256-GCM engine
    OperationService.kt foreground service + progress notification
  assets/images/app_icon.png the one source for every app icon
  test/             unit + widget tests

```

4. Setup, build and verification

4.1 Requirements

- Flutter 3.44 or newer (Dart 3.12+) – matches *pubspec.lock*.
- Android SDK with platform 36, JDK 17, Android Studio or VS Code.
- A device or emulator running Android 8.0 (API 26) or newer.

4.2 First build

```

flutter pub get
flutter run          # debug build on a connected device
flutter build apk --release  # or: flutter build appbundle

```

Release signing: create *android/key.properties* with *storeFile*, *storePassword*, *keyAlias* and *keyPassword*. Without it the release build is signed with the debug key.

4.3 Verify the project

```
flutter pub get
dart fix --apply      # applies automatic lint fixes (e.g. missing const)
flutter analyze      # expected: no errors
flutter test          # unit + widget tests
```

4.4 If the analyzer reports something

- **Info-level lints** (`prefer_const_constructors`, `prefer_final_locals`) are fixed by `dart fix --apply`.
- **An error:** the message names the file and line. The most likely kind, given the code could not be compiled during the upgrade, is a small signature mismatch with a package version – compare with the package source for the version in `pubspec.lock`.
- Run `dart format lib test` to normalise formatting.

4.5 Code generation

None. The project deliberately uses no `build_runner`, `freezed`, `json_serializable` or `drift`. Models are hand-written immutable classes with `copyWith`; the database uses `sqlite` with SQL in `app_database.dart`; strings live in `lib/l10n/app_localizations.dart` (no ARB/gen-l10n). If you introduce a generator later, add `build_runner` to `dev_dependencies` and run `dart run build_runner build --delete-conflicting-outputs`.

4.6 Dependencies

Package	Used for
flutter_riverpod 2.6	State management and dependency injection
go_router 16	Routing, bottom-tab shell (<code>StatefulShellRoute</code>)
sqlite, shared_preferences	Database and settings
flutter_secure_storage, local_auth	Keystore-backed secrets, fingerprint prompts
pointycastle, crypto	Pure-Dart AES-GCM / PBKDF2 fallback, hashes
archive	ZIP/TAR/GZ/BZ2/XZ create and extract
permission_handler, device_info_plus	All-files access, SDK level
video_player + chewie, just_audio, pdfx, photo_view	Built-in viewers
open_file, share_plus	Open with / share
path, path_provider, mime, uuid, intl, logger, collection	Utilities
androidx.core, androidx.appcompat (Gradle)	Notifications; AppCompatActivity theme for the biometric prompt

5. Key modules

5.1 File listing and the folder cache

`FileSystemService.list()` sends one closure to a worker isolate (built in a static method so it captures only the path and a flag). The worker lists the folder and stats every entry with synchronous I/O and returns plain `FileEntry` objects; giant folders are handed to the UI in slices. `BrowserViewModel` first paints the last listing from `DirectoryCache` (LRU of 32 folders), then replaces it with the fresh one, then fills in folder item counts with one call to `childCounts()`. Finished operations invalidate the affected folders.

5.2 The operation queue

`OperationsController` keeps a serial queue of `FileOperations` (copy, move, trash, delete, restore, compress, extract, encrypt, decrypt) with pause, resume and cancel. `OperationRunner` executes one operation:

- Move: same-volume items are renamed first (instant, no pre-scan). Only the rest is flattened (in an isolate) for exact totals and copied.
- Copy: files up to 16 MB use native `File.copy`; larger files are copied in 4 MB slices so pause and cancel react immediately. Partial files are always removed.
- Conflicts ask the UI (Replace / Skip / Keep both, optionally for all).
- Progress is throttled to ~10 updates/s with a rolling 3-second speed average; the card and the notification show percent, bytes, files, speed and time left.
- At the end (also on failure/cancel) added and removed paths are sent to `MediaStore` (`syncMedia`), so galleries stay correct.

5.3 Secure Folder (vault)

Item	Design
Keys	A random 256-bit master key encrypts every file. It is wrapped with a key derived from the 6-digit PIN (PBKDF2-HMAC-SHA256, 120,000 rounds) and stored in <code>meta.json</code> together with a verifier. The master key exists in memory only while the vault is open.
File format (.fv)	magic "FVLT" version u8 chunkSize u32 noncePrefix[8] plainSize u64, then per chunk: ciphertext 16-byte GCM tag. Chunk <i>i</i> uses nonce = prefix u32(<i>i</i>) and authenticates header u32(<i>i</i>) lastFlag, so chunks cannot be reordered, dropped or truncated.
Engines	Kotlin <code>VaultCryptoChannel</code> (javax.crypto AES/GCM, hardware accelerated, 4 MB chunks) on Android; pure-Dart pointycastle engine (1 MB chunks) in tests and as fallback. Both write and read the identical format; readers take the chunk size from the header.
Adding files	Folders are expanded into their files (each keeps its original path). Each file is encrypted to " <code>uuid.fv.part</code> ", renamed to ".fv", recorded, then the original is deleted, its <code>MediaStore</code> row removed and its cached thumbnail evicted. Failures roll back so nothing is lost.
Moving out	Decrypts into the original folder, recreating it if needed (Downloads as fallback), then scans it into <code>MediaStore</code> .

Item	Design
Locking	Locks whenever the vault screen is closed and (by default) as soon as the app goes to the background; viewers opened from the vault are closed. Queued transfers carry their own key copy, which is wiped afterwards.

5.4 App Lock

AppLockService stores only a salt and a PBKDF2 hash of the app PIN in secure storage and counts wrong attempts (5 → 30-second cooldown). *AppLockController* locks on cold start, and on return from the background once the configured delay has passed. The lock screen is drawn above the router in *MaterialApp.builder*, blocks semantics and focus underneath and swallows the back gesture. *LifecycleGuard* stops the system fingerprint dialog from counting as "leaving the app". While App Lock is on, `FLAG_SECURE` hides the app in recents.

5.5 Global progress card

The card is hosted in its own Overlay above every route, so it is visible on tabs, pushed screens and inside the Secure Folder. It hides while a dialog or bottom sheet is open (a navigator observer counts popup routes) and sits above bottom bars that register their height with *ReserveBottomSpace*.

5.6 Theme and design tokens

AppColors defines a small palette: warm paper neutrals, the ink-blue brand from the icon and the amber keyhole accent. *FvTokens* (a ThemeExtension) adds *tonal/onTonal* (soft icon tiles and pills), *vault/onVault* (deep ink used by everything private), chip, border, shadow and success colours. Content on brand fills is always white in both themes; text and outlined buttons are white in dark mode. Never hard-code colours in widgets.

6. Android integration

Topic	Details
Permissions	<code>MANAGE_EXTERNAL_STORAGE</code> on Android 11+ (declare "File manager" in the Play Console), legacy storage below; notifications; foreground service (dataSync); biometrics.
Channels	<i>com.filevault/platform</i> : <code>getStorageVolumes</code> , <code>getVideoThumbnail</code> , <code>getApkInfo</code> , <code>scanMedia</code> , <code>syncMedia</code> , <code>setSecureWindow</code> , operation notification calls, <code>openAllFilesAccessSettings</code> . <i>com.filevault/crypto</i> : <code>encryptFile</code> , <code>decryptFile</code> , <code>cancel</code> ; progress events back to Dart.
MediaStore safety	Rows are deleted only for paths that no longer exist, because on Android 11+ deleting a row also deletes the file.
Icons and splash	Generated from <code>assets/images/app_icon.png</code> : legacy and round launcher icons, adaptive icon (gradient background + foreground + monochrome for themed icons), Android 12 splash icon, pre-12 launch drawable. Re-generate all densities when the artwork changes.
Themes	<code>LaunchTheme/NormalTheme</code> are AppCompat themes – required by the biometric prompt on Android 8.

7. Testing

Test file	Covers
file_system_service_test	listing, hidden files, create/rename/duplicate, small and large copies, cancelled copies leave nothing, flatten, stats, hashes, child counts, existence batch, .nomedia
vault_crypto_test	key wrapping, PBKDF2, multi-chunk round trip, empty file, tamper detection
natural_sort_test, sort_spec_test	natural ordering, folders first, size/date sorting
directory_cache_test, file_clipboard_test	LRU behaviour; copy/cut/clear
operations_controller_test	queue state, progress maths, time left
archive_service_test, search_filter_test, file_utils_test	archives, search filters, classification
widget_test, widget_smoke_test	dark-mode button colours, app icon, file rows, empty states

8. How to extend

Add a screen

- Create *features/<name>/* with a view and a Notifier view model.
- Add a route name in *app_routes.dart* and a GoRoute in *app_router.dart* (full-screen routes use *parentNavigatorKey: rootNavigatorKey*).
- Open folders with *openFolder()* from *open_file.dart* – it picks push or go correctly.

Add a file operation

- Add a value to *OperationType*, a case in *OperationRunner.run*, an enqueue method in *OperationsController*, a verb in *operationVerb* and a done text in the progress panel.
- Report progress through the ticker and note added/removed paths for MediaStore.

Add a setting

- Field + *copyWith* in *AppSettings*, key in *AppConstants*, load/save in *SettingsRepositoryImpl*, setter in *SettingsController*, row in *SettingsView*.

Add a language

- Subclass *AppLocalizations*, override the getters, add the locale to *supportedLocales* and return the subclass from the delegate for that locale.

Change the colours

- Edit *AppColors* and the *FvTokens.light/dark* constants. Keep white content on brand fills (WCAG AA needs $\geq 4.5:1$) and update *fv_splash_background* in the Android colour resources to match the new surface colour.

9. Security model in brief

- Everything stays on the device; the app has no internet permission.
- Vault content is encrypted at rest with AES-256-GCM inside app-private storage, which other apps, galleries and USB/PC connections cannot read. Backups and device transfer exclude app data.
- The PIN is never stored. Five wrong attempts trigger a 30-second cooldown (vault and App Lock).
- Fingerprint unlock of the vault keeps a copy of the master key in Keystore-backed secure storage; it is removed when fingerprint unlock is turned off. Users who want the strongest setting should use the PIN only.
- Limits: uninstalling the app or clearing its data destroys the vault; a forgotten vault PIN cannot be recovered (reset erases the vault). Decrypted previews live in the app cache only while the vault screen is open.

10. Getting started

On first launch FileVault asks for **All files access** (Android 11 and newer) or storage permission (older Android). Tap **Grant access**, switch on FileVault in the system screen and come back. Without it FileVault still opens but can only see its own folders; you can grant access later in **Settings → All files access**.

The splash screen stays up until your storage, categories and recent files are ready, so the first screen you see is complete. If App Lock is on, you are asked for your app PIN or fingerprint next.

Navigation

Tab	What it is for
Home	Free space at a glance, categories, recent files, Secure Folder, quick-access folders, shortcuts to Favorites, Tags, Trash and Activity.
Browse	Every folder on internal storage, SD cards and USB drives.
Search	Instant search across the whole phone with type, size and date filters.
Settings	Appearance, browsing, trash, App Lock, Secure Folder, about.

11. Home

- **Storage card** – how much is free, and a bar showing what fills your phone (images, videos, apps...). An SD card or USB drive appears as a small chip in the corner; tap it to browse.
- **Free up space** opens the storage analyser (large files, duplicates, junk, empty folders).
- **Categories** – tap to see every image, video, document... on the phone. The first time, tap **Scan** to build the index that powers categories and search.
- **Recent files**, **Secure Folder** card (shows Locked/Open), **Quick access** folders and the tool shortcuts. Pull down to refresh.

12. Browsing and managing files

To...	Do this
Open a file or folder	Tap it. Images, videos, audio, PDFs, text, archives and APKs open in FileVault; other types offer "Open with".
See all actions for one item	Tap the three-dot button on its row.
Select several items	Long-press one, then tap others. Use Select all / Invert in the top bar.
Copy or move (paste anywhere)	Select → Copy or Move (or the three-dot menu → Copy / Cut). Walk to the destination folder and tap Paste here in the bar at the bottom. The close (X) button cancels.
Copy or move with a folder picker	Select → More → Copy to... / Move to...
Rename many files at once	Select them → More → Rename all... , type a name. Files become "Name 1.jpg", "Name 2.jpg"... in the current sort order (numbers are zero-padded when there are 10 or more); extensions are kept.

To...	Do this
Delete	Select → Delete moves to Trash. When it finishes, the progress card offers Undo for a few seconds. More → Delete permanently skips the trash.
Create	Tap + for a new folder, new text file or archive.
Hide a folder from galleries	Three-dot menu on a folder → Hide from gallery (adds a .nomedia marker). Choose Show in gallery to undo. The files stay where they are – use the Secure Folder to truly hide them.
Change view and order	Top bar: grid/list and sort (name, size, date, type, folders first, hidden files).
Filter the current folder	Tap the search icon in the folder's top bar.

Progress

Every copy, move, compress, extract and Secure Folder transfer shows a progress card at the bottom of whatever screen you are on, plus a notification: percentage, bytes and files done, transfer speed and time left. Copies and moves can be paused; anything can be cancelled. Tap the arrow to shrink the card to one line. Operations keep running when you leave the app.

13. Search, categories, archives and cleanup

- **Search** finds files anywhere instantly once the index is built; filter by type, size and date. **Settings** → **Rebuild search index** refreshes it.
- **Archives**: open ZIP, TAR, TAR.GZ/TGZ, GZ, BZ2 and XZ to browse and extract selected entries; create ZIP (optionally password-protected), TAR or TAR.GZ archives from a selection.
- **Storage analyser** (Home → Free up space): largest folders and files, duplicate finder (SHA-256), junk and cache cleaner, empty folders, one-tap "Clean everything safe".
- **Favorites, Tags, Recents, Activity** – from Home; tags are coloured labels you can put on any file.

14. Trash

Deleted items go to the Trash first. Open it from Home. Restore puts items back in their original folder (or Downloads if that folder is gone). Items are erased automatically after the period chosen in **Settings** → **Auto-clean trash** (7, 30 or 60 days, or never); expired items are purged when the app starts.

15. Secure Folder guide

The Secure Folder encrypts files on your phone and removes the originals. Encrypted files cannot be seen by galleries, other apps or a computer connected by USB – only inside FileVault after you unlock it.

Set it up

- Home → Secure Folder (or Settings → Secure Folder).
- Choose a 6-digit PIN and enter it again. Optionally allow fingerprint unlock.
- Write the PIN down somewhere safe: **it cannot be recovered**.

Add files, videos and folders

- Inside the Secure Folder tap **Add files**. Tap files to select them; **long-press a folder** to secure everything inside it.
- Or, anywhere in the app: select items (or use the three-dot menu of one) → **Move to Secure Folder**. You will be asked for your PIN/fingerprint first.
- Videos of any size are supported. Progress, speed and time left are shown while they are encrypted; you can leave the Secure Folder or the app meanwhile.

Open and move out

- Tap a file to view it. Large files show a progress bar while they are decrypted; Cancel stops it.
- Long-press to select → **Move out of vault** restores files to their original folder (recreated if needed) and they appear in your gallery again.
- Delete inside the vault erases files permanently – there is no trash for encrypted files.

When it asks for the PIN again

- Every time you open the Secure Folder.
- When FileVault goes to the background (default "Immediately"; change in Settings → Auto-lock). Any file you were viewing from the vault is closed.
- Lock it yourself at any time with the lock icon.

Fingerprint, PIN and reset

- Turn fingerprint on or off, or change the PIN, from the three-dot menu inside the open Secure Folder.
- Five wrong PINs lock input for 30 seconds.
- **Forgot PIN?** on the lock screen resets the vault – this permanently destroys every file in it.
- Screenshots and screen recording are blocked while the Secure Folder is open.

16. App Lock guide

App Lock protects the whole app with its own 6-digit PIN (it can differ from the Secure Folder PIN).

- **Turn on:** Settings → Security → App lock → choose and confirm a PIN, optionally allow fingerprint.
- **Lock after leaving:** Immediately (default), 30 seconds, 1 minute or 5 minutes in the background.
- **Change app PIN:** enter the current PIN, then the new one twice.
- **Turn off:** switch it off and confirm with your app PIN.
- While App Lock is on, FileVault is hidden in the recent-apps screen and screenshots are blocked. Five wrong PINs lock input for 30 seconds.
- Forgot the app PIN? Clear FileVault's data in Android settings. **Warning:** this also erases the Secure Folder.

17. Settings reference

Setting	Effect
All files access	Shows whether FileVault may manage all files; opens Android settings.
Theme / Accent colour	System, light or dark; five accent colours.
Default layout, Sort order	How folders open by default.
Show hidden files	Show names starting with a dot.
Confirm file deletion	Ask before moving to Trash (Undo is offered either way).
Auto-clean trash	Erase trashed items after 7/30/60 days or never.
App lock, Unlock with fingerprint, Lock after leaving, Change app PIN	See chapter 16.
Block screenshots	Blocks screenshots and hides the app in recents everywhere in FileVault.
Secure Folder, fingerprint, Auto-lock	See chapter 15.
Rebuild search index	Re-scans storage for search and categories.
Privacy & encryption, Open source licenses	Information.

18. FAQ and troubleshooting

Categories show "—" and search finds nothing.

The index has not been built yet. Tap Scan on Home or Settings → Rebuild search index.

A folder says "Restricted by Android".

Android/data and Android/obb cannot be read by any app on Android 11+. Use a computer for those.

A file I moved into the Secure Folder still shows in my gallery.

FileVault removes it from Android's media index immediately. Some gallery apps keep their own thumbnail cache – refresh the gallery or clear its cache once.

Fingerprint for the Secure Folder can't be turned on from Settings.

It needs your PIN first: open the Secure Folder and use its three-dot menu.

Moving a big folder on the same storage is instant, but to the SD card it takes time.

Within one storage a move is a rename. Between storages every byte must be copied, then the originals deleted.

A copy failed with "Not enough free space".

Free space on the destination and retry; the partial file is removed automatically.

The progress card covers something.

Tap the arrow to shrink it to one line. It hides automatically while dialogs and sheets are open.

I forgot my Secure Folder PIN.

It cannot be recovered. "Forgot PIN?" resets the vault and destroys its files.

Do my files leave the phone?

No. FileVault has no internet permission and is excluded from cloud backup.

Will the Secure Folder survive an uninstall?

No. Move files out before uninstalling or clearing data.